$$\mathrm{d}\boldsymbol{p} = -\nabla U(\boldsymbol{q})\mathrm{d}t - \boldsymbol{\Phi}(\boldsymbol{\xi}, z) - \boldsymbol{\Gamma}\boldsymbol{p}\mathrm{d}t + \sqrt{2\mathrm{k_B T}}\boldsymbol{M}^{1/2}\boldsymbol{\Gamma}^{1/2}\mathrm{d}\mathbf{W}, \tag{8.23}$$

$$\mathrm{d}\boldsymbol{\xi} = \boldsymbol{g}(\boldsymbol{\xi}, \boldsymbol{q}, \boldsymbol{p})\mathrm{d}t - \hat{\boldsymbol{\Gamma}}\boldsymbol{\xi}\mathrm{d}t + \sqrt{2\mathrm{k_B T}}\hat{\boldsymbol{M}}^{1/2}\hat{\boldsymbol{\Gamma}}^{1/2}\mathrm{d}\hat{\mathbf{W}}. \tag{8.24}$$

Here $\boldsymbol{\Gamma}, \hat{\boldsymbol{\Gamma}}$ are symmetric, square matrices of appropriate dimension, $\mathbf{W}$ is a vector of $\mathrm{N_c}$ Wiener processes, and $\hat{\mathbf{W}}$ is a vector of m Wiener processes, all independent of each other. We assume that the deterministic thermostat represented by the $\boldsymbol{\Phi}$ and $\boldsymbol{g}$ terms preserves the extended distribution $\hat{\rho}(\boldsymbol{\xi})\rho_\beta$, where $\hat{\rho}(\boldsymbol{\xi}) = \exp\left(-\beta\boldsymbol{\xi}^T\hat{\boldsymbol{M}}^{-1}\boldsymbol{\xi}/2\right)$. The matrix $\hat{\boldsymbol{M}}$ is symmetric and positive definite; one could take it to be diagonal.

The additivity property of thermostats mentioned above holds also for stochastic thermostats, since the Fokker-Planck equation is constructed from linear operations on the vector fields involved, thus it is enough to require that the newly introduced terms satisfy appropriate fluctuation-dissipation relationships so that they are compatible with the extended Gibbs distribution of the deterministic part. That is, we require

$$\mathrm{d}\boldsymbol{p} = -\boldsymbol{\Gamma}\boldsymbol{p}\mathrm{d}t + \sqrt{2\mathrm{k_B}T}\boldsymbol{M}^{1/2}\boldsymbol{\Gamma}^{1/2}\mathrm{d}\mathbf{W}$$

to preserve the distribution with density $\bar{\rho} = \exp\left(-\beta\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2\right)$, and

$$\mathrm{d}\boldsymbol{\xi} = -\hat{\boldsymbol{\Gamma}}\boldsymbol{\xi}\mathrm{d}t + \sqrt{2\mathrm{k_B T}}\hat{\boldsymbol{M}}^{1/2}\hat{\boldsymbol{\Gamma}}^{1/2}\mathrm{d}\hat{\mathbf{W}}$$

to preserve $\hat{\rho}$. These two problems are essentially the same. Considering the former, the stationary Fokker-Planck equation reads

$$\underbrace{\nabla_{\boldsymbol{p}} \cdot (\boldsymbol{\Gamma}\boldsymbol{p}\bar{\rho})}_{\text{drift}} + \underbrace{\beta^{-1}\mathrm{Tr}\left(\boldsymbol{\Gamma}^{1/2}\boldsymbol{M}^{1/2}\nabla_{\boldsymbol{p}}^2\bar{\rho}\boldsymbol{M}^{1/2}\boldsymbol{\Gamma}^{1/2}\right)}_{\text{diffusion}} = 0.$$

The drift term here equates to

$$\mathrm{Tr}(\boldsymbol{\Gamma})\bar{\rho} + (-\beta)\boldsymbol{p}^T\boldsymbol{\Gamma}\boldsymbol{M}^{-1}\boldsymbol{p}\bar{\rho}.$$

For the diffusion term, we calculate

$$\nabla\bar{\rho} = -\beta\bar{\rho}\boldsymbol{M}^{-1}\boldsymbol{p},$$

thus

$$\nabla^2\bar{\rho} = \bar{\rho}[\beta^2\boldsymbol{M}^{-1}\boldsymbol{p}\boldsymbol{p}^T\boldsymbol{M}^{-1} - \beta\boldsymbol{M}^{-1}],$$

so the diffusion term is

$$\beta^{-1}\mathrm{Tr}\left(\boldsymbol{\Gamma}^{1/2}\boldsymbol{M}^{1/2}[\beta\boldsymbol{M}^{-1}\boldsymbol{p}\boldsymbol{p}^{T}\boldsymbol{M}^{-1}-\boldsymbol{M}^{-1}]\boldsymbol{M}^{1/2}\boldsymbol{\Gamma}^{1/2}\right)\bar{\rho}=-\mathrm{drift},$$

where we have used the cyclic property of the trace (and the symmetry of $\boldsymbol{\Gamma}$) to reduce the expression. The validity of the term involving $\boldsymbol{\xi}$ is easily verified by a similar argument.

Thus we know that the Gibbs distribution will be invariant. It remains to prove that the SDE (8.22)–(8.24) is ergodic. Certainly this would be expected in the case where a full Langevin dynamics contribution is present ($\boldsymbol{\Gamma}$ positive definite), but as we shall see next there are cases where noise only contacts the auxiliary degrees of freedom but the system is nonetheless ergodic.

We state here a useful observation that allows to simplify the commutator calculations in the case of an SDE of the following type:

$$\mathrm{d}\boldsymbol{z}=\boldsymbol{F}(\boldsymbol{z})\mathrm{d}t+\xi\boldsymbol{G}(\boldsymbol{z})\mathrm{d}t, \tag{8.25}$$

$$\mathrm{d}\xi=g(\boldsymbol{z})\mathrm{d}t-\gamma\xi\mathrm{d}t+\sigma\mathrm{dW}, \tag{8.26}$$

where $\boldsymbol{z}\in\mathbb{R}^m$, $\boldsymbol{F},\boldsymbol{G}:\mathbb{R}^m\to\mathbb{R}^m$.

Proposition 8.2 *If the Lie algebra generated by $\boldsymbol{F}$ and $\boldsymbol{G}$ in (8.25) spans $\mathbb{R}^m$, then the Hörmander condition holds for the system (8.25)–(8.26).*

Proof First observe that the premise holds if and only if the Lie algebra of the vectors in $\mathbb{R}^{m+1}$ defined by

$$\tilde{\boldsymbol{F}}=\begin{bmatrix}\boldsymbol{F}\\0\end{bmatrix},\qquad\tilde{\boldsymbol{G}}=\begin{bmatrix}\boldsymbol{G}\\0\end{bmatrix},$$

together with $\boldsymbol{e}_{m+1}$ spans $\mathbb{R}^{m+1}$. Thus the problem is to show that we can generate $\tilde{\boldsymbol{F}}$ and $\tilde{\boldsymbol{G}}$ from the vector fields that define (8.25)–(8.26), which are

$$\boldsymbol{b}_0=\begin{bmatrix}\boldsymbol{F}+\xi\boldsymbol{G}\\g-\gamma\xi\end{bmatrix},\qquad\boldsymbol{b}_1=\begin{bmatrix}\boldsymbol{0}\\\sigma\end{bmatrix}.$$

Computing the commutator gives

$$\boldsymbol{c}=[\boldsymbol{b}_0,\boldsymbol{b}_1]=\begin{bmatrix}-\sigma\boldsymbol{G}\\\sigma\gamma\end{bmatrix},$$

hence

$$\tilde{\boldsymbol{G}}=\frac{-1}{\sigma}\left[\boldsymbol{c}-\gamma\boldsymbol{b}_1\right].$$